

- (a) 10 (b) 12
(c) 16 (d) 18
23. $2 - 2 + 2 - 2 + \dots$ 101 terms =? (P.C.S., 2008)
(a) -2 (b) 0
(c) 2 (d) None of these
24. 98th term of the infinite series 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, is (M.C.A., 2005)
(a) 1 (b) 2
(c) 3 (d) 4
25. If x, y, z be the digits of a number beginning from the left, the number is
(a) xyz (b) $x + 10y + 100z$
(c) $10x + y + 100z$ (d) $100x + 10y + z$
26. If x, y, z and w be the digits of a number beginning from the left, the number is
(a) $xyzw$
(b) $wzyx$
(c) $x + 10y + 100z + 1000w$
(d) $10^3x + 10^2y + 10z + w$
27. If n and p are both odd numbers, which of the following is an even number?
(a) $n + p$ (b) $n + p + 1$
(c) $np + 2$ (d) np
28. For the integer n , if n^3 is odd, then which of the following statements are true?
I. n is odd. II. n^2 is odd.
III. n^2 is even.
(a) I only (b) II only
(c) I and II only (d) I and III only
29. If $(n - 1)$ is an odd number, what are the two other odd numbers nearest to it?
(a) $n, n - 1$ (b) $n, n - 2$
(c) $n - 3, n + 1$ (d) $n - 3, n + 5$
30. Which of the following is always odd?
(a) Sum of two odd numbers
(b) Difference of two odd numbers
(c) Product of two odd numbers
(d) None of these
31. If x is an odd integer, then which of the following is true?
(a) $5x - 2$ is even (b) $5x^2 + 2$ is odd
(c) $5x^2 + 3$ is odd (d) None of these
32. If a and b are two numbers such that $ab = 0$, then (R.R.B., 2006)
(a) $a = 0$ and $b = 0$ (b) $a = 0$ or $b = 0$ or both
(c) $a = 0$ and $b \neq 0$ (d) $b = 0$ and $a \neq 0$
33. If A, B, C, D are numbers in increasing order and D, B, E are numbers in decreasing order, then which one of the following sequences need neither be in a decreasing nor in an increasing order?
(a) E, C, D (b) E, B, C
(c) D, B, A (d) A, E, C
34. If m, n, o, p and q are integers, then $m(n + o)(p - q)$ must be even when which of the following is even?
(a) m (b) p
(c) $m + n$ (d) $n + p$
35. If n is a negative number, then which of the following is the least?
(a) 0 (b) $-n$
(c) $2n$ (d) n^2
36. If $x - y = 8$, then which of the following must be true?
I. Both x and y are positive.
II. If x is positive, y must be positive.
III. If x is negative, y must be negative.
(a) I only (b) II only
(c) I and II (d) III only
37. If x and y are negative, then which of the following statements is/are always true?
I. $x + y$ is positive.
II. xy is positive.
III. $x - y$ is positive.
(a) I only (b) II only
(c) III only (d) I and III only
38. If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following is/are true?
I. n is odd. II. n is prime.
III. n is a perfect square.
(a) I only (b) I and II only
(c) I and III only (d) None of these
39. If $x = \frac{2}{5}y + 3$, how does y change when x increases from 1 to 2?
(a) y increases from -5 to $-\frac{5}{2}$
(b) y increases from $\frac{2}{5}$ to 5
(c) y increases from $\frac{5}{2}$ to 5
(d) y decreases from -5 to $-\frac{5}{2}$
40. If x is a rational number and y is an irrational number, then
(a) both $x + y$ and xy are necessarily rational
(b) both $x + y$ and xy are necessarily irrational
(c) xy is necessarily irrational, but $x + y$ can be either rational or irrational
(d) $x + y$ is necessarily irrational, but xy can be either rational or irrational
41. The difference between the square of any two consecutive integers is equal to
(a) sum of two numbers
(b) difference of two numbers
(c) an even number
(d) product of two numbers